



DATA DUEL

Question vs Answer Challenge

Ask Smart. Think Fast. Answer Right.

Activity introduction & rules
for Data Analytics / Data Science classrooms

Excel | Power BI | SQL | Python | Aptitude

01 INTRODUCTION

What is Data Duel?

Data Duel is a team-based Data Analytics knowledge challenge. One team frames the questions, the other team answers them, and both are evaluated every round.

It tests

Conceptual understanding

Question-framing ability

Technical knowledge

Business thinking

Communication skills

Problem-solving

Ability to explain concepts

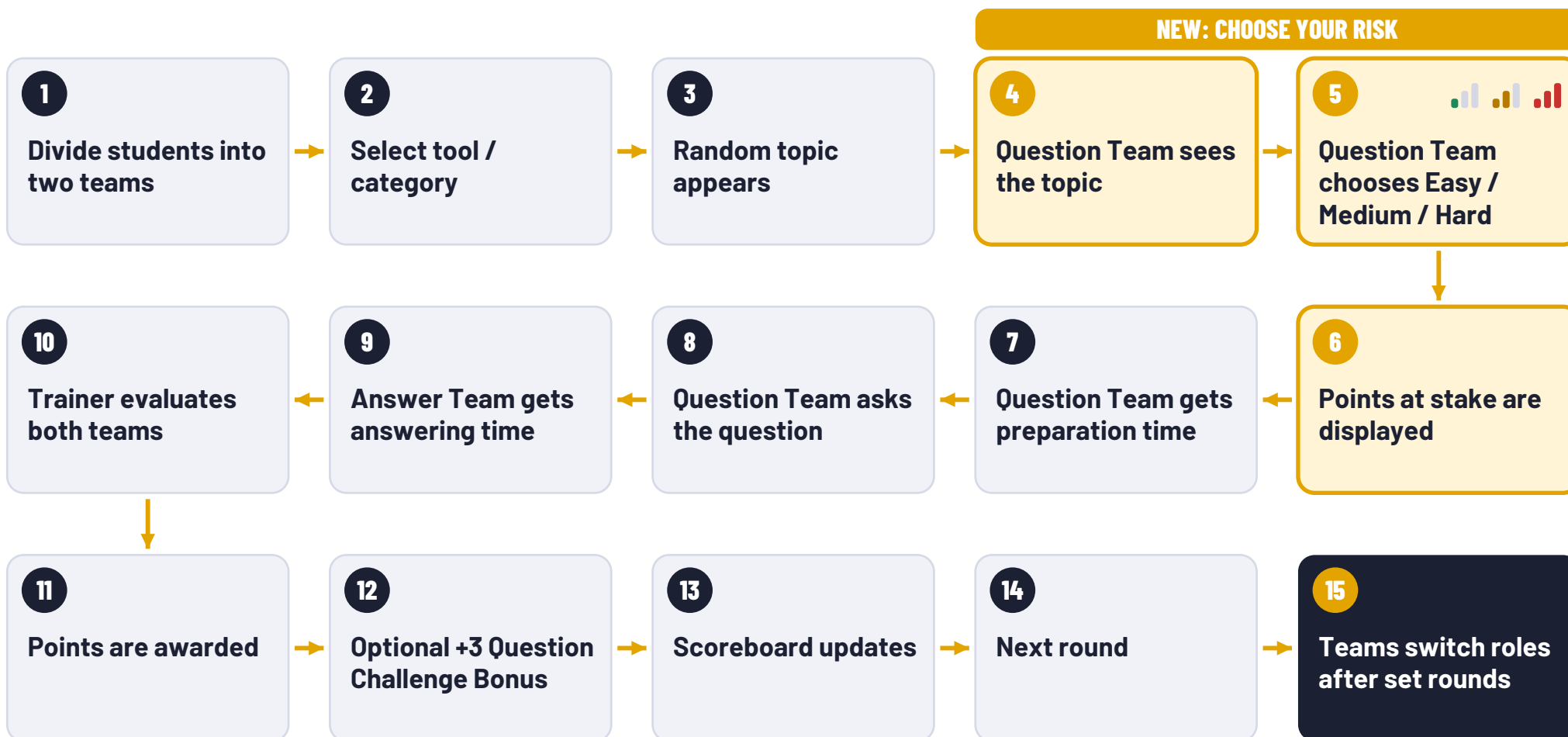
Thinking under time pressure

More than a mock interview

In a traditional mock interview, students only answer. In Data Duel, students also **create meaningful technical questions**, **choose how much to risk** on them, and must be able to answer their own question.

02 ACTIVITY FLOW

How the activity works



TOPIC FIRST, RISK SECOND

The Question Team picks Easy, Medium or Hard only after the topic appears. The choice is locked before the question is asked.

03 WHO DOES WHAT

Team roles



QUESTION TEAM

Frames the challenge

- After the topic appears, choose Easy, Medium or Hard before asking.
- Create a question based on the displayed topic.
- Make sure the question is technically valid.
- Make sure the question is easy to understand.
- Test knowledge or application, not trivia.
- The student asking must know the correct answer.
- Never create impossible or misleading questions on purpose.



ANSWER TEAM

Solves the challenge

- Listen carefully to the full question.
- Answer within the given time.
- Explain the reasoning wherever required.
- Give an example wherever possible.
- Use correct technical terminology.

THE KEY RULE

If a Question Team member asks a question, they must know the correct answer to that question.

04 DIFFICULTY-BASED QUESTION SCORING

Choose your risk

Topic is revealed



Question Team chooses the difficulty



Locked before the question is asked

 **EASY****2 POINTS**

Maximum for the Question Team

Test basic concepts and definitions.**Lower risk, lower reward.****Example question (SQL: Window Functions)***"What is a window function in SQL?"* **MEDIUM****4 POINTS**

Maximum for the Question Team

Test understanding, comparison and practical application.**Balanced risk and reward.****Example question (SQL: Window Functions)***"What is the difference between ROW_NUMBER(), RANK() and DENSE_RANK()?"* **HARD****6 POINTS**

Maximum for the Question Team

Test reasoning, troubleshooting, business scenarios and deeper technical understanding.**Higher risk, higher reward.****Example question (SQL: Window Functions)***"A Sales table has Customer_ID, Sale_Date and Amount. How would you find each customer's second-highest transaction while keeping their other records? Which window function would you use, and why?"***Choose your difficulty wisely.**

The harder the question, the greater the potential Question Team score.
The Answer Team can always earn up to 6, whatever risk is chosen.

05 POINTS PER ROUND

Scoring system

QUESTION TEAM		Maximum depends on the risk chosen		
Criteria		 EASY	 MEDIUM	 HARD
Relevant to the topic		+1	+1	+1
Technically correct		+1	+1	+1
Clearly framed		+1	+1	+1
Tests understanding / application		-	+1	+1
Requires reasoning / business scenario		-	-	+1
Strong technical depth		-	-	+1
Maximum		2	4	6

Easy merges "technically correct" and "clearly framed" into one point.

ANSWER TEAM		6 POINTS
Criteria		Points
Correct answer		+3
Explanation / reasoning		+1
Example / application		+1
Technical terminology		+1
Maximum		6

Always out of 6, whatever risk the Question Team chose.

MAXIMUM PER ROUND

Question + Answer, before bonus

 **8** 2 + 6

 **10** 4 + 6

 **12** 6 + 6

+3
bonus on top

06 +3 BONUS

Question Challenge Bonus

1 Answer Team misses

The Answer Team gives an incorrect or incomplete answer.

**2 Trainer challenges**

The trainer turns to the student who asked the question.

**3 Question Master answers**

If the Question Team student answers correctly, the team earns the bonus.

THE TRAINER ASKS

"Question Master, can you answer your own question?"

+3**Question Master answers correctly**

Added **on top of** the Easy, Medium or Hard question score. It reinforces the rule: **never ask a question you don't know the answer to.**

0**Both teams answer incorrectly**

No bonus points are awarded. The trainer shares the correct answer before moving to the next round.

07 ONE ROUND, START TO FINISH

Worked example round

1 TOPIC REVEALED

SQL: WINDOW FUNCTIONS

2 QUESTION TEAM CHOOSES

HARD: 6 POINTS

3 THE QUESTION

"A sales table contains multiple transactions for every customer. How would you identify the second-highest transaction amount for each customer while retaining all customer records?"

4 TRAINER EVALUATES THE QUESTION

5 / 6☒ Relevant to the topic☒ Technically correct☐ Clearly framed☒ Tests understanding / application☒ Requires reasoning / business scenario☒ Strong technical depth

"Clearly framed" is missed: "retaining all customer records" could mean several different outputs.

Outcome A: Answer Team answers well

Correct answer, explanation and terminology, but no example.

QUESTION TEAM

5

out of 6 (Hard)

ANSWER TEAM

5

out of 6

Outcome B: Answer Team misses

The trainer asks: "Question Master, can you answer your own question?" The student answers correctly.

QUESTION TEAM

5 + 3 = 8

Hard score + bonus

ANSWER TEAM

0

incorrect answer

The +3 bonus is added on top of the chosen difficulty score.

The trainer gives the correct answer if both teams miss.

08 PLAY FAIR

Activity rules

01 The displayed topic must be followed.

02 The Question Team must ask a relevant question.

03 The Question Team member must know the answer.

04 Difficulty is chosen after the topic appears and cannot change once the question is asked.

05 No Google, ChatGPT, mobile phones, notes or external help during the round.

06 No repeated questions.

07 Questions must be technically valid.

08 The Answer Team must answer within the allotted time.

09 Answers should include reasoning wherever applicable.

10 The Trainer / Judge has the final decision.

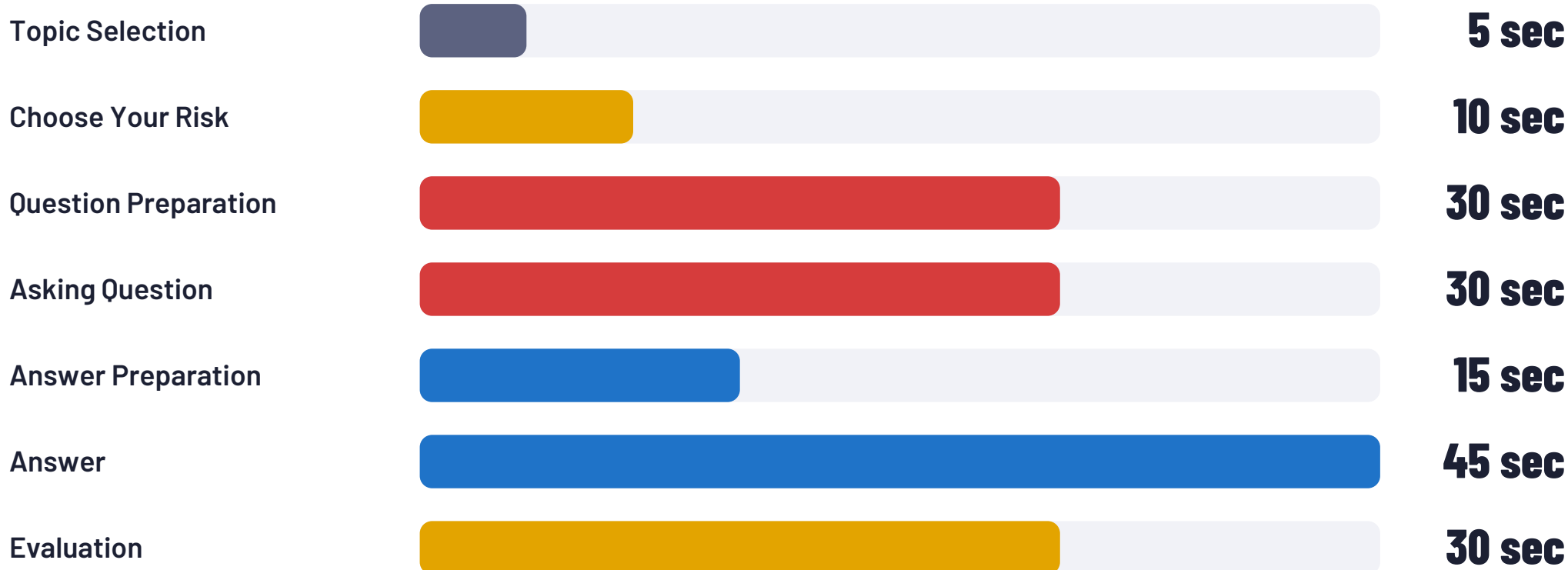
11 Teams must switch roles after the specified number of rounds.

12 Respectful communication is mandatory.

13 No mocking or interrupting another student.

09 THE CLOCK

Recommended timing

**TOTAL, APPROXIMATELY****2-3 minutes per round**

These are recommendations. The trainer can change every timer in the Topic Shuffler's Settings panel.

10 ASK BETTER QUESTIONS

Question quality examples

Topic on screen:

SQL JOIN

WEAK**“What is INNER JOIN?”**

Tests recall of a definition. Easy to memorise, says little about skill.

BETTER**“What is the difference between INNER JOIN and LEFT JOIN?”**

Tests understanding by comparing two related ideas.

STRONG**“A company has a Customers table and a Transactions table. How would you identify customers who have registered but have never made a transaction?”**

Tests application: the student must pick LEFT JOIN with an IS NULL check (or NOT EXISTS) to solve a real business problem.

Strong questions test application, not just memorisation. Aim for a short scenario with a clear, checkable answer.

11 OPTIONAL TWISTS

Special rounds



Follow-Up Attack

After the answer, the Question Team may ask one follow-up question on the same topic.



Business Scenario

A business situation is displayed instead of a technical topic. Teams think like analysts.



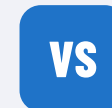
Rapid Fire

The Answer Team gets only 15 seconds to answer.



Find the Error

Faulty SQL, Python, DAX or Excel logic is displayed. Spot the mistake and correct it.



Concept Clash

Compare two related concepts: what each does and when to use which.

Concept Clash examples

WHERE vs HAVING

SUM vs SUMX

VLOOKUP vs XLOOKUP

List vs Tuple

Mean vs Median

loc vs iloc

RANK vs DENSE_RANK

Import vs DirectQuery

UNION vs UNION ALL

Permutation vs Combination

The Question Team still chooses Easy, Medium or Hard in every special round, after the topic or scenario appears.

12 EVERYONE CAN SHINE

Winning & recognition

Recognise individuals, not only the winning team.



Best Question Master



Best Answer Master



Best Technical Question



Best Business Scenario



Fastest Thinker



Best Follow-Up Question



Best Data Analyst

Write winners' names on the board
or in the Shuffler's Final Score.

TEAM SCOREBOARD

TEAM	QUESTION PTS	ANSWER PTS	BONUS	TOTAL
Team Q				
Team A				

13 TRAINER QUICK GUIDE

Running it with the Topic Shuffler

1 Set up

Open the Topic Shuffler on the projector. In Settings, enter team names, total rounds, how often teams switch roles, and timer lengths.

2 Choose tools

Pick one, several or all tools and, optionally, a special round. There is no difficulty setting: the Question Team decides that.

3 Shuffle

Press Shuffle topic. Topics don't repeat until every topic in the selected tools has been used.

4 Choose your risk

The Question Team taps Easy (2), Medium (4) or Hard (6), or you press 1, 2 or 3. The choice locks for the round and the preparation timer starts.

5 Play

When preparation ends, the answer timer is ready. Use the follow-up timer for Follow-Up Attack.

6 Score

Tick each criterion earned. Only the criteria for the chosen risk appear, so scores can't exceed 2 / 4 / 6, the Answer Team's 6, or the +3 bonus.

7 Rotate

The screen prompts you when it's time to switch roles. Press Switch teams; the change applies from the next round.

8 Finish

Open Final score to show the result with each team's question, answer and bonus points, and type in the award winners.

SHORTCUTS**S**

Shuffle / next round

1 2 3

Easy / Medium / Hard

T

Start or pause timer

W

Switch teams

APPENDIX A EXCEL, POWER BI, SQL

Topic bank

Tiers show each topic's natural level. In play, the Question Team chooses Easy, Medium or Hard.

EXCEL

37 topics

● Easy

VLOOKUP, HLOOKUP, SUMIF, COUNTIF, IF, Sorting, Conditional Formatting, Text Functions, Date Functions, Logical Functions, Excel Charts, Data Validation, Duplicate Handling, Absolute vs Relative References, Excel Tables

● Medium

XLOOKUP, INDEX MATCH, SUMIFS, COUNTIFS, IFS, Nested IF, Pivot Tables, Pivot Charts, Advanced Filter, Lookup Functions, Data Cleaning, Missing Values, Goal Seek, Named Ranges, TEXTJOIN & CONCAT, Slicers & Timelines

● Hard

What-If Analysis, Scenario Manager, Power Query, Dynamic Arrays, Business Analytics, LET & LAMBDA

POWER BI

39 topics

● Easy

Power Query, Data Cleaning, Fact Table, Dimension Table, Relationships, Measures, Calculated Columns, KPIs, Tooltips, Slicers, Dashboard Design, Import Mode, Visual, Page & Report Filters

● Medium

Data Transformation, Merge Queries, Append Queries, Data Modeling, Star Schema, Cardinality, Calculated Tables, FILTER, ALL, SUMX, COUNTX, Time Intelligence, DATEADD, SAMEPERIODLASTYEAR, PREVIOUSMONTH, Drill Through, Bookmarks, DirectQuery, Date Table

● Hard

Query Folding, CALCULATE, REMOVEFILTERS, KEEPFILTERS, Row-Level Security, Filter Context vs Row Context, Cross-Filter Direction

SQL

41 topics

● Easy

SELECT, WHERE, DISTINCT, ORDER BY, LIMIT, OFFSET, Aggregate Functions, INNER JOIN, UNION, String Functions, Primary Key, Foreign Key

● Medium

GROUP BY, HAVING, LEFT JOIN, RIGHT JOIN, SELF JOIN, CROSS JOIN, UNION ALL, CASE WHEN, Subqueries, CTE, NULL Handling, COALESCE, Date Functions, Normalization, Views, Finding Duplicate Rows

● Hard

GROUP BY + HAVING, Correlated Subqueries, Recursive CTE, Window Functions, ROW_NUMBER, RANK, DENSE_RANK, LEAD, LAG, Indexes, Transactions, Nth Highest Salary, Running Totals

APPENDIX B PYTHON, APTITUDE, CONCEPT CLASH

Topic bank

Tiers show each topic's natural level. In play, the Question Team chooses Easy, Medium or Hard.

PYTHON

39 topics

● Easy

Variables, Data Types, Lists, Tuples, Sets, Dictionaries, Functions, Loops, Conditional Statements, Modules, Series, DataFrame, value_counts

● Medium

Lambda Functions, List Comprehension, Dictionary Comprehension, Exception Handling, File Handling, NumPy, Pandas, loc, iloc, concat, fillna, dropna, Datetime, Data Cleaning, Matplotlib, Seaborn, Mutable vs Immutable, *args and **kwargs

● Hard

groupby, merge, pivot_table, apply, map, Lambda with Pandas, EDA, Outlier Detection

APTITUDE

26 topics

● Easy

Percentages, Averages, Ratios, Simple Interest, Number Systems, Series, Coding-Decoding, Direction Sense

● Medium

Profit and Loss, Time and Work, Time Speed Distance, Compound Interest, Logical Reasoning, Data Interpretation, Blood Relations, Basic Statistics, Mixtures & Alligations, Boats and Streams

● Hard

Probability, Permutations, Combinations, Syllogisms, Puzzles, Quantitative Reasoning, Clocks and Calendars, Seating Arrangement

CONCEPT CLASH PAIRS

Excel

VLOOKUP vs XLOOKUP, COUNT vs COUNTA, SUMIF vs SUMIFS, Absolute vs Relative Reference, Pivot Table vs Power Query

Power BI

SUM vs SUMX, Calculated Column vs Measure, Import vs DirectQuery, Merge vs Append Queries, ALL vs REMOVEFILTERS, Star vs Snowflake Schema

SQL

WHERE vs HAVING, INNER JOIN vs LEFT JOIN, UNION vs UNION ALL, RANK vs DENSE_RANK, DELETE vs TRUNCATE vs DROP, Subquery vs CTE

Python

List vs Tuple, loc vs iloc, merge vs concat, apply vs map, dropna vs fillna, List vs Set

Aptitude

Mean vs Median, Permutation vs Combination, Simple vs Compound Interest, Correlation vs Causation, Variance vs Standard Deviation

182 topics in total. The Shuffler also includes business scenarios and Find-the-Error snippets.